

# Simulating Ground States and Dynamics of Small Quantum Many-Body Systems Using DMRG

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# Motivation

- Calculating the ground states of quantum many-body systems is a fundamental problem in condensed matter physics and quantum chemistry. For one-dimensional strongly correlated systems, exact diagonalization (ED) can provide benchmark results, but its computational complexity grows exponentially with system size, severely limiting the size of systems that can be studied.
- Since its introduction by Steven R. White in 1992, the Density Matrix Renormalization Group (DMRG) has become one of the most important numerical tools for solving one-dimensional strongly correlated systems. White's groundbreaking work shifted the approach to quantum many-body problems from traditional "diagonalizing large matrices" to "constructing and optimizing the reduced density matrix between the system and the environment." By retaining the eigenstates corresponding to the largest eigenvalues, DMRG achieves a significant reduction in computational complexity with minimal loss of accuracy.

# Motivation

- In the following two decades, DMRG underwent several major developments. Schollwöck and others unified DMRG with the matrix product state (MPS) formalism, revealing that DMRG is essentially a variational optimization process of MPS. This perspective not only clarified the mathematical structure of the algorithm but also laid the theoretical foundation for subsequent improvements. Based on this, researchers have proposed optimization strategies such as dynamically controlling bond dimensions, adding noise perturbations, and measuring multi-point correlation functions, which have significantly improved the convergence speed and accuracy of DMRG.

# Objective

- Understand the physical principles of DMRG and MPS/MPO
- Implement a DMRG algorithm based on existing code or literature
- Compare DMRG performance with exact diagonalization (TEBD/iTEBD)
- Simulate ground state properties of one-dimensional hydrogen chains
- Investigate optimal bond length, Peierls dimerization, and energy comparison with isolated  $\text{H}_2$  molecules

# Project Outline (20/04-15/06, 9 Weeks)

## 1 Literature Review & Project Design (Wk1)

- Understand the principles of the DMRG algorithm
- Identify specific computable problems
- Complete project design, determine the assignment for the first-state.

## 2 Code Implementation (Wk2-3)

- Set up a Linux virtual machine, create the experimental environment, and install required libraries.
- Carry out code development in parallel according to division of labor:

# Project Outline (20/04-15/06, 9 Weeks)

## ② Code Implementation (Wk2-3)

- Set up a Linux virtual machine, create the experimental environment, and install required libraries.

- Carry out code development in parallel according to division of labor:

*STEP1* (JJ.Xu) :Implement DMRG in Python to solve the ground state of the 1D Heisenberg spin chain. Explain the computational principles of DMRG with reference to the code.

*STEP2* (JY.Wang) :Use iTEBD and DMRG separately to solve the ground state of the same system. Use TEBD and tDMRG separately to handle time evolution problems.

*STEP3* (JT.Wang&SY.Li) :Import block2 to compute the ground state energy of the hydrogen molecule. Calculate the optimal bond length for equally spaced hydrogen chains with different numbers of atoms under equal atom distance. Compute the lowest-energy configuration of the H chain when dimerization is taken into account.

### **3 Code Optimization & Integration (Wk4)**

- Cross-run other team members' code to ensure correctness and stability of calculations.
- Analyze experimental results and discuss the underlying physical meaning
- Refine the code, add comments and result analysis, and integrate everything into a single Jupyter Notebook file

### **4 Midterm Report Preparation & Paper Writing (Wk5-7)**

- Adjust algorithm details and paper structure based on feedback
- Prepare in-class presentation
- Divide the paper writing tasks among team members and draft the main sections respectively

### **5 Results Presentation & Paper Revision (Wk8-9)**

- Oral presentation
- Finalize the paper
- Compile all research results

# References

## 1 DMRG Algorithm

- Chan, G. K.-L., Keselman, A., Nakatani, N., & White, S. R. (2016). Matrix product operators, matrix product states, and ab initio density matrix renormalization group algorithms. *Journal of Chemical Physics*, \*145\*(1), 014102. <https://doi.org/10.1063/1.4955108>
- Keller, S., Dolfi, M., Troyer, M., & Reiher, M. (2015). An efficient matrix product operator representation of the quantum chemical Hamiltonian. *Journal of Chemical Physics*, \*143\*(24), 244118. <https://doi.org/10.1063/1.4939000>
- Wouters, S., & Van Neck, D. (2014). The density matrix renormalization group for ab initio quantum chemistry. *European Physical Journal D*, \*68\*(9), 272. <https://doi.org/10.1140/epjd/e2014-50500-1>

## 2 Quantum Chemistry and Hydrogen Chain Structure

- Kadzielawa, A. P., Biborski, A., & Spalek, J. (2015). Discontinuous transition of molecular-hydrogen chain to the quasiautomic state: Combined exact diagonalization and ab initio approach. Physical Review B, \*92\*(16), 165101.  
<https://doi.org/10.1103/PhysRevB.92.161101>
- Sawaya, R. C., & White, S. R. (2022). Constructing Hubbard models for the hydrogen chain using sliced-basis density matrix renormalization group.  
\*Physical Review B, 105\*(4), 045145.  
<https://doi.org/10.1103/PhysRevB.105.045145>

# Deliverables

- 1. **A complete Jupyter Notebook**, including DMRG algorithm implementation (Step 1), Performance comparison between DMRG and other algorithms (Step 2), DMRG simulation of the hydrogen chain (Step 3). A dmrg\_project conda environment package is also provided. It can be installed and run on Linux, with figures saved automatically.
- 2. **An oral presentation**, covering the experimental background, objectives, project methodology, computational process, and result analysis. It will also include code demonstration and discussion of the results.
- 3. **A written report** detailing the physical principles of DMRG and MPS/MPO, the algorithm implementation and performance on different systems, the suitable physical problems and advantages of DMRG, and the ground state calculation results for the 1D hydrogen chain, including the optimal bond length, Peierls dimerization effect, energy comparison with isolated  $H_2$  molecules, and the trend toward independent molecules in the thermodynamic limit.

# Assessment Criteria

- 1. Depth of understanding of DMRG and MPS/MPO
- 2. Quality and clarity of the written report and presentation.
- 3. Accuracy and completeness of the simulation results and analysis.
- 4. Creativity and critical thinking in discussion.